

LAB-WEEK-2 (8 puzzle problem using dfs)

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# 8 Puzzle using DFS

def dfs(start, goal):
    stack = [(start, [])]

    while stack:
        state, path = stack.pop()

        if state == goal:
            return path + [state]

        if len(path) >= 6:
            continue

        zero = state.index(0)

        row = zero // 3
        col = zero % 3

        moves = []

        if col > 0:
            moves.append(zero - 1)
        if row > 0:
            moves.append(zero - 3)

        if row < 2:
            moves.append(zero + 3)
        if col < 2:
            moves.append(zero + 1)

        for move in reversed(moves):

            new_state = list(state)
            new_state[zero], new_state[move] = \
                new_state[move], new_state[zero]

            new_state = tuple(new_state)

            stack.append((new_state, path + [state]))

    return None
```

```
print("name : Hrishikesh shet 1WN24CS120")

print("Enter 9 numbers (0 represents blank):")
start = tuple(map(int, input().split()))

goal = (
    1, 2, 3,
    4, 5, 6,
    7, 0, 8
)

solution = dfs(start, goal)

if solution:

    print("\nSolution found!")
    print("Number of moves:", len(solution) - 1)

    for state in solution:
        print()
        print(state[0], state[1], state[2])
        print(state[3], state[4], state[5])
        print(state[6], state[7], state[8])

else:
    print("No solution found within 6 moves.")
```

OUTPUT:

```
*** name : Hrishikesh shet 1wN24CS120
Enter 9 numbers (0 represents blank):
1 2 3 0 4 6 7 5 8

Solution found!
Number of moves: 6

1 2 3
0 4 6
7 5 8

0 2 3
1 4 6
7 5 8

1 2 3
0 4 6
7 5 8

0 2 3
1 4 6
7 5 8

1 2 3
0 4 6
7 5 8

1 2 3
4 0 6
7 5 8

1 2 3
4 5 6
7 0 8
```